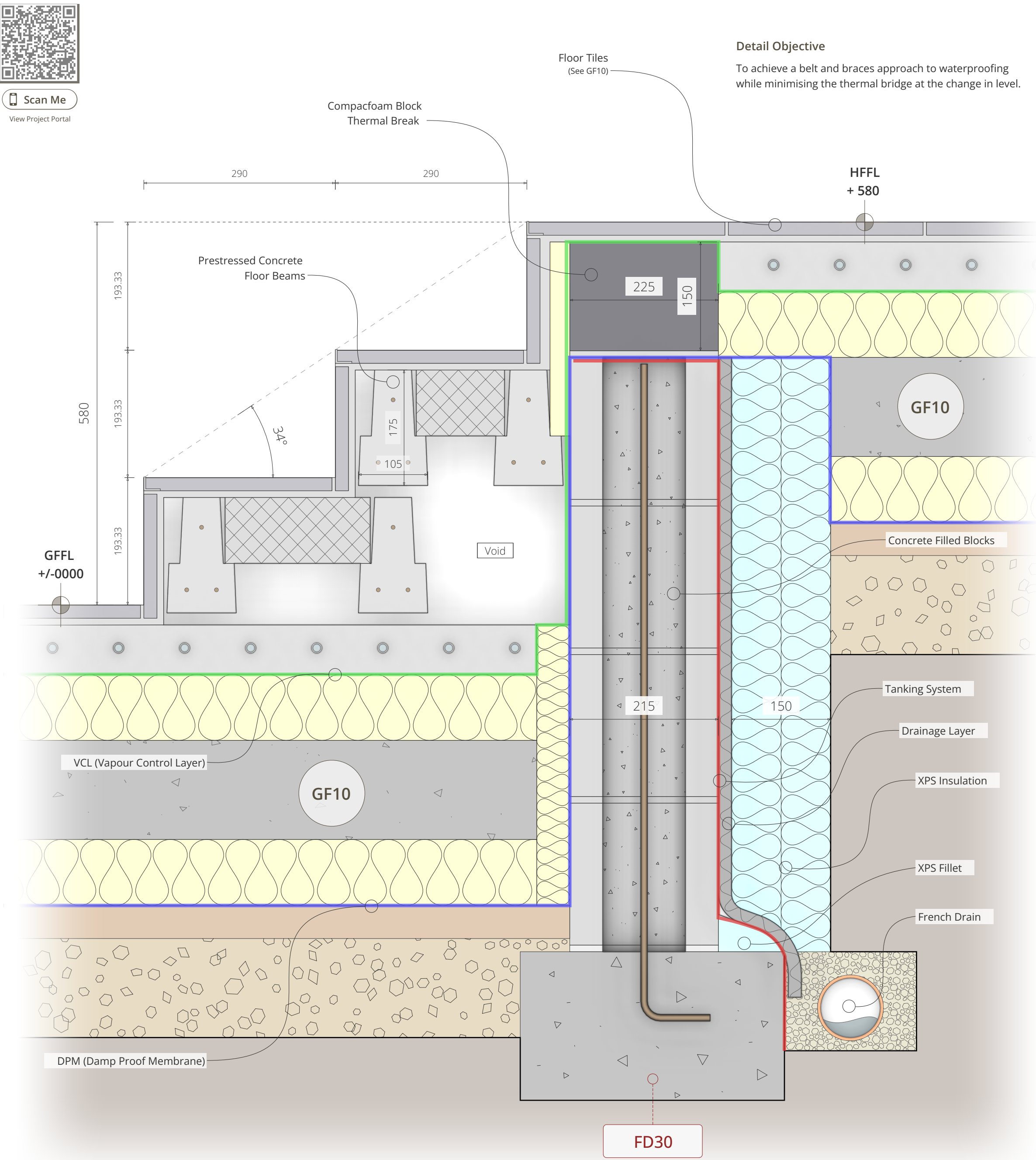




SP10 - Steps & Level Transition Detail

Building Control Approval Submission

This drawing is submitted for Building Control approval. Construction must not commence until the Inspector sanctions the proposed structural details, thermal performance, and moisture integrity. The Principal Contractor is fully responsible for coordinating all statutory inspections to verify strict compliance with the Building Regulations 2010.

Element	Specifications
External Tanking	Tanking to be "Type A". Apply self-adhesive bituminous waterproofing membrane (e.g. Visqueen Self Adhesive Membrane or Hydrobond) directly to the primed external face of the blockwork. Primer must be applied to dry clean blockwork before peeling and sticking the membrane vertically with minimum 150mm overlaps. Roll firmly to ensure full adhesion. Surfaces must be free of dust and snots. This forms the primary Type A Barrier Protection to BS 8102:2022. IMPORTANT: Terminate this layer by lapping it 215mm horizontally across the very top flat surface of the concrete retaining wall (capping the wall) before the Compacfoam block is installed.
External Drainage Layer	Install a Geocomposite Drainage & Protection Sheet (e.g. Delta Terraxx or Visqueen Protect&Drain) directly over the tanking membrane. Fix mechanically at the very top (above the waterproofing line) or adhere with compatible bitumen dabs. The dimples must face inwards towards the wall to create an air gap for water depressurisation. Ensure the bottom edge connects directly to the land drain/french drain pipe.
Retaining Wall Insulation	150mm High-Performance Extruded Polystyrene (XPS) such as Kingspan GreenGuard GG300 or Styrofoam LB. Install tightly butted against the drainage/tanking layer on the earth side. To match the thermal resistance of the main floor's 100mm PIR (0.022 W/mK) this layer MUST be 150mm thick (XPS is 0.034 W/mK). Install as 2x 75mm or 3x 50mm staggered layers to prevent thermal bypassing. CRITICAL: Do not use foil-faced PIR (Celotex) in this location; it will absorb groundwater and fail over time.
Retaining Wall Structure	215mm Hollow Dense Concrete Blocks (7.3N or higher) such as Thermalite Stepoc or standard dense hollows. Install H12 or H16 vertical rebar (or as per Structural Engineer's schedule) into the hollow cores anchored into the foundation. Fill cores with C30/35 structural concrete (NOT lean mix) and vibrate to remove air pockets. This monolithic structure retains the higher ground level.
Top Edge Thermal Break	h150mm x w225mm Compacfoam (Grade CF200) Structural Thermoplastic Block. Bed this block using a structural adhesive directly onto the tanked top surface of the concrete retaining wall. This block acts as a structural edge restraint for the screed and It ensures the screed does not touch the cold retaining wall below.
Step Structure (Suspended)	Prestressed Concrete Floor Beams (T-Beams) & Infill Blocks (e.g. Bison/Forterra). Cut beams to span between the lower bearing and the retaining wall niche to form a suspended step structure. Grout all joints with 1:4 cement:sand mix. Ensure minimum 100mm bearing on the wall structures. Install 10mm Structural Thermal Break Pads (e.g. Armatherm FRR or Farrat TBK) directly under the ends of the prestressed concrete floor beams where they bear onto the retaining wall. These high-compressive strength pads isolate the cold concrete wall from the floor beams preventing a cold bridge tracking into the floor structure.
Step Finish & Decoupling	Final tiling to Client choice. CRITICAL: Install a Movement/Decoupling Mat (e.g. Schluter-DITRA or similar) over the step tread and the transition between the screed and Compacfoam block before tiling. This isolates the tiles from differential movement between the plastic Compacfoam and the concrete screed preventing hairline cracks in the grout or tiles.
DPM Continuity	The blue line denotes a1200 Gauge (300 micron) Polythene DPM. The DPM must form a continuous unbroken barrier from the lower level to the upper level. Install as follows: 1. Lay DPM under the lower floor. 2. Run DPM vertically up the internal face of the retaining wall. 3. Lap DPM up and OVER the top of the retaining wall ABOVE the tanking. 4. Seal/Lap onto the upper floor DPM. This creates a continuous sealed tray that prevents damp rising through the step structure or wall connection.
Under-Step Void	The void beneath the suspended steps is sealed and small. Responsibility for ventilation or gas protection (Radon/Methane) lies with the Building Control Officer's site discretion. If required by inspector install periscopic vents; otherwise assume sealed void condition acceptable due to continuous membrane protection below (DPM & VCL).
Design Rationale	1. Thermal Continuity: We introduced Armatherm pads under the beams and a Compacfoam block at the screed edge to ensure no part of the warm floor structure touches the cold retaining wall. 2. Waterproofing Logic: The Red Line tanking caps the wall externally. The Blue Line DPM runs internally and laps over the thermal break creating a double-layer protection at the critical junction. 3. Structural Stability: Suspended steps avoid settlement; the decoupling mat prevents finish cracking at the material transition.
Standards	BS 8102:2022 (Waterproofing), BS EN 1996 (Masonry), BS EN 13164 (XPS Insulation), Part L 2021 (Conservation of Fuel & Power), BS 5250 (Control of Condensation).